BERNARDO BARRERA

bbarrera@bu.edu ⋄ Google Scholar

EDUCATION

Ph.D. in Physics *August 2022–June 2028 (Expected)*
Boston University
Supervised by Anushya Chandran and Anatoli Polkovnikov

B.S. in Engineering Physics *August 2017–June 2022*
Instituto Tecnológico y de Estudios Superiores de Monterrey

RESEARCH EXPERIENCE

Non-Adiabatic Dynamics and Mixed Quantum-Classical Methods *January 2023–Present*
BU Condensed Matter Theory Group

- Developed a mixed quantum-classical framework extending the Born-Oppenheimer approximation, with applications to quantum chemistry and condensed matter systems. Published in *PNAS*.
- Carried out computational benchmarks using numerical simulations in Python and MATLAB of quantum systems with sparse matrices, Krylov methods, and semi-classical techniques.

Topology in Circuit QED *May 2025–Present*
In collaboration with the Kollár Group at UMD

- Theory student in an experimental project realizing a topological photon pump using a transmon qubit coupled to a microwave cavity.
- Demonstrated topological pumping from vacuum up to 7 photons, and the generation of non-classical cavity states.
- Developed the theoretical model for the experiment, and carried out numerical simulations in Python of driven qubit-cavity dynamics using Lindblad master equations to understand experimental data.

Dynamics in Billiards *January 2022–August 2022*
Photonics and Mathematical Optics Group, Tecnológico de Monterrey

- Derived new analytical expressions for trajectory winding numbers and periods in an integrable billiard, and validated the results numerically in MATLAB. Published in *Physical Review E*.

Gravitational Lensing Models *June 2020–August 2020*
University of Minnesota

- Developed and numerically optimized flexible mass models in Fortran to reproduce observational data from a gravitationally lensed quasar. Published in *The Open Journal of Astrophysics*.

PUBLICATIONS

Barrera, B., Verma, N., Queiroz, R., Polkovnikov, A., Chandran, A. (2026). Kinematic Quantum Geometry. *Manuscript in preparation.*

Yue, Q., **Barrera, B.**, Ritter, M., Long, D. M., Lane, D. A., Chandran, A., & Kollár, A. J. (2026). Realization of a quantum topological photon pump. *arXiv:2608.00162.*

Barrera, B., Arovas, D. P., Chandran, A., & Polkovnikov, A. (2026). The moving Born–Oppenheimer approximation. *Proceedings of the National Academy of Sciences*, 123(7), e2507816123.

Barrera, B., Ruz-Cuen, J. P., & Gutiérrez-Vega, J. C. (2023). Elliptic billiard with harmonic potential: Classical description. *Physical Review E*, 108(3), 034205.

Barrera, B., Williams, L. L. R., Coles, J. P., & Denzel, P. (2021). Bridging the gap between simply parametrized and free-form pixelated models of galaxy lenses: The case of WFI 2033-4723 Quad. *The Open Journal of Astrophysics*, 4(1).

SKILLS

Physics	Quantum many-body theory and simulation, open quantum systems, phase space methods, mixed quantum-classical dynamics, non-adiabatic dynamics, circuit QED
Programming	Python, MATLAB
Languages	Spanish (native), English (fluent), Portuguese (intermediate)

AWARDS

Goldhaber Prize for early excellence in first year of graduate school *2024*
Boston University

Teaching Fellow of the Year *2023*
Boston University

Best GPA of the School of Engineering and Sciences *2022*
Instituto Tecnológico y de Estudios Superiores de Monterrey

1st **Place Physics & Math ANFEI Mexican National Science Competition** *2019*
Instituto Tecnológico y de Estudios Superiores de Monterrey

Academic Excellence Scholarship *2017–2022*
Instituto Tecnológico y de Estudios Superiores de Monterrey

TALKS AND POSTERS

- Poster at the 2026 Molecular Interactions and Dynamics Gordon Research Conference.
  Title: *Geometric Phases and Forces in Chiral Pseudospin Textures.*
- Invited talk at ETH Zurich ITP, 2025.
  Title: *The Moving Born–Oppenheimer Approximation.*
- Invited talk at the 2025 Chaos and Geometry in Quantum Many-Body Systems Conference, Ettore Majorana Foundation and Centre for Scientific Culture.
  Title: *The Moving Born–Oppenheimer Approximation.*
- Contributed talks at the 2024, 2025, 2026 APS Global Physics Summits.